

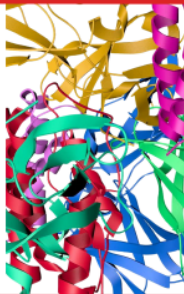
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A comparison of amphibian (*Xenopus laevis*) peritoneal- and bone marrow–derived mast cells

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Abstract

Mast cells are tissue-resident granulocytes that are often viewed through a negative lens due to their roles in allergies and hypersensitivity. While central to these conditions, mast cells have broader roles in vertebrate immune systems that are often overlooked. Whereas most cell research to date has been conducted in mammalian models, mast-like cells are found across vertebrates, presumably with conserved and divergent functions in disparate immune systems. Our past studies defined a role of amphibian (*Xenopus laevis*) mast cells in skin antifungal defenses but stopped short of functionally characterizing these immune effectors. Presently, to further elucidate the biology of amphibian mast cells and to delineate evolutionarily conserved and divergent mast cell characteristics, we developed in vitro *X. laevis* mast cell cultures from peritoneal-derived and bone marrow–derived precursors. The phenotypes, transcriptomes and degranulation capacities were examined and compared between these frog mast cell cultures and to those of frog neutrophils, a distinct granulocyte lineage. Our results indicate that *X. laevis* mast cells are distinct from frog neutrophils and share many canonical features that are associated with mammalian mast cells. Akin to their mammalian counterparts, frog mast cells are capable of degranulation and release potent immune proteins and lipid mediators. Notably, although many characteristics appear to be shared between frog and mammalian mast cells, our findings also indicate unique aspects to these frog leukocytes. Future studies of nonmammalian mast cells will grant unique perspectives of the evolutionary pressures that shape mast cell defenses and associated pathologies.

Keywords: mast cells, comparative immunology, type 2 immunity, *Xenopus laevis*, tissue-resident granulocytes

Introduction

Mast cells were first observed in 1863 by Friedrich von Recklinghausen who described a subset of granulated cells in the mesenteries of frogs.¹ These cells were later independently described by Paul Ehrlich in 1878, who named them “mastzellen” or “well-fed cells,” due to their granular morphology seen by metachromatic staining.² Although mast cells contribute to immune defense, homeostasis, and tissue repair, they are often villainized for their role in type I hypersensitivity and allergies.³ As a result, most studies on mast cell biology have been in the context of allergies and strategies to inhibit mast cell–mediated responses.⁴ Only relatively recently have advances in mast cell biology linked this cell type with broader more integral roles in vertebrate immune systems and host defenses.⁵ Mast cells are particularly important for type 2 immune responses against parasites such as helminths,⁶ but also to certain bacterial,⁷ fungal,⁸ and viral pathogens,⁹ participating in both the initial innate immune responses and engaging the adaptive immune system.^{6–12}

Mast cells strategically reside within barrier and connective tissues, thus representing a critical sentinel leukocyte type evolved to quickly respond to immune assaults.¹³ Mast cells contain acidic granules filled with bioactive mediators, which they can quickly release in response to immune stimuli, allowing for a rapid immune response.³ The activators that initiate mast cell degranulation include, but are not limited to, pathogens, venoms, allergens, and neuropeptides.

Subsequent degranulation leads to the immediate release of diverse mediators, including proteases, histamine, leukotrienes, cytokines, and heparin.¹⁴ Activated mast cells also release de novo effectors including cytokines, chemokines, and lipid mediators that are synthesized and released independently of degranulation.¹⁵ Context-dependent mast cell degranulation may lead to inflammation,¹⁶ tissue remodeling,¹⁷ vascular permeability,¹⁸ and chemotaxis,¹⁹ as well as direct antimicrobial effects.^{14,16–20} The most thoroughly studied mammalian mast cell degranulation pathway is through antigen crosslinking to IgE and signaling through the high-affinity IgE receptor (FcεRI).²¹ FcεRI is a multiprotein complex composed of α and β subunits and signals through the Fcγ chain. The high-affinity receptor binds monomeric IgE, and through antigen crosslinking, many FcεRI complexes aggregate leading to degranulation.²² However, mammalian mast cells can also degranulate independently of FcεRI activation, such as through pattern recognition receptors and Mas-related G protein–coupled receptors X2/B2 (MRGPRX2/B2 found in humans and mice, respectively).^{21,23,24}

Mammalian mast cells are classified by their tissue locations as either connective tissue mast cells (CTMCs) or mucosal mast cells (MMCs).^{14,25} CTMCs are found within the intestinal submucosa, skin, and surrounding blood vessels.^{3,26} MMCs reside in mucosal tissues such as the gastrointestinal tract, and respiratory mucosa.^{26,27} These mast cell subtypes exhibit distinct phenotypes and effector profiles,

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enabling the distinction between CTMCs and MMCs.²⁶ For example, the proteoglycan heparin is present in CTMC granules but absent from MMC granules.²⁸ Additionally, their protease repertoires differ such that CTMCs contain both tryptase and chymase (human connective tissue mast cells), whereas MMCs predominantly contain tryptase (human mucosal mast cells); however, there are many other differences including distinct Toll-like receptor (TLR) and cytokine repertoires.^{26,29,30}

Mast (like) cells are found across vertebrates including birds, amphibians, teleost fish, and even some invertebrates.^{31–34} However, much of what is known about mast cell biology has been informed from mammalian models, with very few functional studies on mast cells in nonmammalian vertebrates, such as amphibians. Notably, amphibians occupy a key evolutionary position between teleost fish and amniotes, providing an opportunity to investigate the evolutionary history of mast cells as well as the conservation and divergence of their functions.³⁵ This also includes the evolution of Ig-mediated mast cell responses, as amphibians lack IgE but instead utilize IgY, which is also found in birds/reptiles and is predicted to be the predecessor of IgG and IgE.³⁶

Past studies of amphibian mast cells noted annual variations in mast cell numbers and visualized mast cells in various tissues, with the greatest number observed in the tongue and a large concentration of mast cells surrounding the nerves of the frog, *Rana esculenta*.^{37,38} Our previous work demonstrated that frog mast cells, akin to their mammalian counterparts, develop in response to the stem/mast cell factor and that these cells are important to amphibian skin defenses against the *Batrachochytrium dendrobatidis* chytrid fungus.³⁹ These studies on amphibian mast cells have paved the way to further characterize these cells but left many questions unanswered regarding the biology and functionality of these amphibian leukocytes. To grant new perspectives on possible frog mast cell subtypes and to begin to delineate the functionalities of these nonmammalian effectors, we developed methods for generating mature *X. laevis* mast cells from distinct precursor sources. We compared the transcriptomes of these mast cells with frog neutrophils and examined mast cell degranulation capacities. Here, we report on both conserved and divergent mast cell functions in a nonmammalian species, providing new insights into the evolution of mast cell biology and the broader immune roles of this relatively overlooked immune cell type.

Methods

Recombinant cytokine production

Recombinant cytokines were produced as previously described.^{39–41} In brief, coding sequences for *X. laevis* stem cell factor (SCF) (NM_001086656) and colony-stimulating factor 3 (CSF3/granulocyte colony-stimulating factor, XM_018234703.2), excluding the signal peptide sequences, were cloned into the pMIB/V5-His A insect expression vector (Invitrogen). The constructs were transfected into Sf9 insect cells using Cellfectin II reagent (Invitrogen), and the cells were cultured in Sf-900 II media until reaching confluency. The cells were pelleted and the supernatants containing the secreted recombinant proteins were collected for purification. Recombinant proteins were purified using Ni-NTA agarose beads (Qiagen) according to the manufacturer's instructions. Protein purity and identity were confirmed by SDS-PAGE

with Coomassie Blue staining and western blot analysis using an anti-V5-HRP antibody (Sigma-Aldrich).

Mast cell and neutrophil cultures

Bone marrow-derived mast cells (BMMCs) and neutrophils were cultured using previously described methods.^{42,43} Adult *X. laevis* were euthanized with 5% tricaine mesylate, followed by decapitation. Both femurs were removed and placed in ice cold amphibian phosphate-buffered saline (A-PBS) (100 mM NaCl₂, 8 mM NaH₂PO₄, 1.5 mM KH₂PO₄, pH = 7.4). Under sterile conditions, individual femur bones were flushed with 10 mL of sterile A-PBS to collect bone marrow cells. Cells were pelleted and resuspended in approximately 200 μ L of complete amphibian medium (C-AM) (200 mL Mammalian serum-free basic media, 60 mL H₂O, 25 mL fetal bovine serum, 100 U/mL penicillin, 100 μ g/mL streptomycin, and 15 μ g/mL gentamycin, pH = 7.4). Red blood cells were removed using a differential density gradient against 51% Percoll. Cells were gently layered over 500 μ L of 51% sterile Percoll and centrifuged at 600 g for 10 min at 4°C. White blood cells were carefully aspirated from the top interface, and residual Percoll was removed by washing the cells in 1 mL of C-AM followed by centrifugation at 600 g for 10 min at 4°C. Live cells were counted using trypan blue exclusion and seeded in 48 well tissue culture plates at 10⁵ cells per well in 250 μ L of C-AM. Cells were maintained at 27°C with 5% CO₂, and cytokines were replenished every 3 d. Cells were treated with 250 μ g/mL recombinant SCF (rSCF) in order to generate BMMCs, or with 250 μ g/mL recombinant granulocyte colony-stimulating factor for neutrophil cultures. BMC cultures were maintained for 21 d, while neutrophil cultures were maintained for 5 d.

Peritoneal leukocytes were isolated following a modified version of a previously described method.⁴⁴ Adult *X. laevis* were anesthetized in 1% tricaine mesylate. Approximately 10 mL of sterile A-PBS was injected into the peritoneal cavity using 20-gauge needle attached to a 10 mL syringe. Animal abdomens were gently massaged to detach peritoneal cells from tissue. The A-PBS was carefully withdrawn through the same needle into 15 mL conical tubes on ice. Peritoneal leukocytes were pelleted at 600 g for 10 min at 4°C and resuspended in 200 μ L of C-AM. Samples containing red blood cells were processed using a 51% Percoll density gradient centrifugation, as described previously. Live cells were counted by trypan blue exclusion and plated at 10⁵ cells per well in 250 μ L of C-AM. Cells were cultured at 27°C with 5% CO₂ and received cytokine treatments every 3 d. PMCs were treated with 250 μ g/mL rSCF for 21 d to promote mast cell differentiation.

Histology

Metachromatic staining was performed as previously described,³⁹ with minor modifications. BMMCs, PMCs, and neutrophils were spun onto Superfrost Plus positively charged microscope slides (Thermo Fisher Scientific) either using a cytospin centrifuge or by loading cells into cytology chimneys inserted into a swing bucket rotor centrifuge. Cells were spun at 1,000 g for 10 min at 4°C. Spun cells were stained with Giemsa for 10 min at room temperature, rinsed with A-PBS, and mounted with Permount (Fisher Scientific). Slides were sealed with clear nail polish and stored at room temperature until imaging. PMCs and BMMCs were imaged with a black and white camera, with pseudocoloring applied

subsequently, to approximate the appearance of the metachromatic staining as it was observed under standard bright-field microscopy (Fig. S1).

RNA in situ hybridization

In situ hybridization was performed using the RNAscope Multiplex Fluorescent Reagent Kit v2 (ACD Bio) following the manufacturer's protocol with slight modifications. A custom probe targeting the *X. laevis cpa3.L* (carboxypeptidase A3, L homeolog; ACD Bio probe ID: XI-LOC121393816, channel C3) was used. BMMCs, PMCs, and neutrophils were spun onto Superfrost Plus slides as previously described, fixed in 4% paraformaldehyde for 10 min at room temperature followed by 2 washes in A-PBS. Cells were dehydrated in a graded ethanol series (50%, 70%, 100%) for 5 min each and stored in 100% ethanol at -20°C until further processing. Prior to hybridization, slides were air-dried, treated with hydrogen peroxide (ACD Bio) for 10 min at room temperature, and washed twice in A-PBS. Protease III (ACD Bio) was applied for 10 min at room temperature followed by a 2-min A-PBS wash. The *cpa3* probe was warmed to 37°C , applied to the cells, and hybridized for 2 h at 40°C . Posthybridization washes were performed twice for 2 min with RNAscope Wash Buffer (ACD Bio). Sequential amplifications with amplifier 1 (AMP1), AMP2, and AMP3 (ACD Bio) were carried out at 40°C for 30, 30, and 15 min, respectively, with washes in Wash Buffer between steps.

For fluorophore detection, HRP-C3 (ACD Bio) was added and incubated for 15 min at 40°C , followed by 2 Wash Buffer rinses. TSA Vivid 570 (Cy3 channel, ACD Bio) was diluted 1:750 in TSA buffer and incubated on the slides for 30 min at 40°C . HRP signal was then quenched using HRP blocker (ACD Bio) for 15 min at 40°C .

Immunohistochemistry

Immunohistochemistry was performed with modifications of previously described methods.⁴⁰ Fixed cells, that had not undergone RNAscope processing, were permeabilized with ice cold methanol for 5 min. All cells were then blocked in blocking buffer containing 3% bovine serum albumin and 3% normal goat serum in A-PBS for 1 h at room temperature. Cells were incubated with a mouse anti-actin antibody (1:1,500 dilution; MP Biomedicals) for 1 h at room temperature, followed by 2 washes in A-PBS (2 min each) and an incubation with a secondary goat anti-mouse Alexa Fluor 488 (1:6,000 dilution; Invitrogen) for 1 h at room temperature. Cells were stained simultaneously with Avidin-Texas Red (1:1,500 dilution) to visualize heparin within cytoplasmic granules. Nuclear staining was performed using DAPI (1:1,000 dilution; Thermo Fisher Scientific) for 5 min at room temperature, followed by A-PBS washes. Slides were mounted with ProLong Gold Antifade (Thermo Fisher Scientific), coverslipped, and sealed with clear nail polish. Mounted slides were stored in the dark until imaging, which was performed on either a Zeiss Axio Observer inverted fluorescence microscope or a Zeiss LSM 710 confocal microscope.

RNA sequencing and analysis

RNA sequencing was performed as previously described.³⁹ BMMCs, PMCs, and neutrophils were cultured as outlined previously. Cells were purified by fluorescence-activated cell sorting based on established forward and side scatter profiles. Sorted cells were pelleted at 600 g for 10 min at 4°C and

lysed directly in 500 μL of TRIzol reagent (Invitrogen). RNA was extracted according to the manufacturer's protocol. For phase separation, 200 μL of BCP was added to the cell lysate, vortexed vigorously, incubated at room temperature for 2 min, and centrifuged at 12,000 g for 15 min at 4°C . The upper aqueous phase was transferred to a new tube containing 1 μL of glycogen. RNA was precipitated with 500 μL of isopropanol for 15 min at room temperature and pelleted by centrifugation at 12,000 g for 10 min at 4°C . The RNA pellet was washed with 70% ice cold ethanol and centrifuged at 9,000 g for 7 min at 4°C . Ethanol was removed and the RNA was resuspended in approximately 20 μL of molecular-grade water, followed by incubation at 55°C for 10 min. Purified RNA was flash frozen in liquid nitrogen and stored at -80°C until further processing.

RNA quality and concentration were assessed using an Agilent TapeStation and Qubit fluorometer, respectively. High-quality RNA samples were further processed by Azenta Life Sciences for library preparation and sequencing. Libraries were prepared by PolyA selection to enrich for messenger RNA, followed by fragmentation and cDNA synthesis. Sequencing was performed on an Illumina HiSeq platform to generate 150 bp paired-end reads, with an average depth of 20 million reads per sample. Differential gene expression was analyzed using the DESeq2 package. Genes with fewer than 10 total counts across all samples were excluded to remove lowly expressed transcripts. Genes with an adjusted *P* value of <0.05 were considered differentially expressed. The top 50 most significant differentially expressed genes were log2-transformed and visualized as a heatmap using the pheatmap R package.

To assess the immunomodulatory molecules produced or associated with mast cells and neutrophils, normalized RNA sequencing read counts for selected genes of interest were compared across the different cell cultures. To allow for log transformation and prevent undefined values from zero counts, all normalized read counts were increased by 1 prior to analysis. Differential expression of specific genes between PMCs, BMMCs, and neutrophils was assessed using pairwise Wilcoxon rank-sum tests. *P* values were adjusted for multiple comparisons using the false discovery rate method, with significance defined as an adjusted *P* value <0.05 .

Degranulation assays

X. laevis mast cell degranulation assays were optimized from standard methods employed for mammalian mast cells.^{45–48} In brief, BMMCs and PMCs were cultured as described previously, washed, and resuspended in modified Tyrode's buffer (100 mM NaCl, 5 mM KCl, 0.9 mM CaCl_2 , 1 mM MgCl_2 , 20 mM HEPES, and 5.6 mM glucose, pH 7.4). Cell counts were determined, and 5.0×10^5 cells were plated in a 96-well tissue culture plate. Cells were stimulated for 1 h at 27°C with 500 nM calcium ionophore A23187 (Sigma-Aldrich), or an equivalent volume of DMSO for the control treatment. Following stimulation, cells were pelleted at 600 g for 10 min, and supernatants were transferred to a new 96-well plate for enzymatic activity assays. Tryptase activity was measured by adding BAPNA (Sigma-Aldrich) to a final concentration of 0.4 mM. Reactions were incubated for 90 min in the dark at 27°C , and absorbance was measured at 410 nm using a Synergy HTX plate reader (BioTek). For β -hexosaminidase activity, PNAG was resuspended at 3.5 mg/mL in citrate buffer (40 mM citric acid, 20 mM

Na₂HPO₄, pH 2.5). The PNAG solution was added to supernatants to achieve a final concentration of 2.3 mg/mL and incubated for 90 min at 27°C in the dark. Reactions were terminated by the addition of 50 µL of glycine buffer (0.4 mM glycine, pH 10.7), and absorbance was read at 410 nm. Absorbance values were normalized to the respective DMSO controls for each animal. Statistical significance was assessed using a paired Wilcoxon signed rank test, with $P < 0.05$ considered significant.

Proteomic and cysteinyl leukotriene analysis of mast cell degranulates

BMMCs and PMCs were stimulated with 500 nM A23187 as described previously. For downstream proteomic and cysteinyl leukotriene analyses, degranulates from approximately 2.0×10^6 cells were collected and pooled from 6 individual animals. Supernatants were filtered using Pierce Spin Cups with 0.4 µm cellulose acetate filters (Thermo Fisher Scientific) to remove any cells. Filtered degranulates were immediately flash frozen and stored at -80°C until analysis.

Proteins present in PMC and BMMC degranulates were identified by mass spectrometry, following previously described methodologies.⁴⁹ Supernatants were resuspended in SDS buffer (5% SDS, 50 mM TEABC, and 20 mM DTT), then boiled at 95°C for 10 min. After cooling to room temperature, iodoacetamide was added to a final concentration of 40 mM, and samples were incubated in the dark for 30 min. The samples were centrifuged at 13,000 g for 8 min, and the supernatants were transferred to S-Trap columns (ProtiFi). Proteins were digested overnight at 37°C using sequencing-grade Lys-C/trypsin (Promega). Digested peptides were eluted from the columns and dried using a SpeedVac (Fisher Scientific). Peptides were resuspended in 0.1% formic acid and loaded onto a C18 trap column (Waters Acquity; Cat# 186007496) at a flow rate of 10 µL/min for 4 min. Peptide separation was performed on an analytical column (Waters Acquity; Cat# 186007490) maintained at 40°C with a flow rate of 350 nL/min, using a 150-min gradient of buffer A (2% acetonitrile, 0.1% formic acid) and buffer B (0.1% formic acid in acetonitrile). Mass spectrometry was conducted using a nanoAcquity UPLC system (Waters) coupled to an Orbitrap Fusion Lumos mass spectrometer (Thermo Fisher Scientific). Protein identification was performed with the Sequest HT algorithm in the Proteome Discoverer platform (Thermo Fisher Scientific), using the *X. laevis* proteome database from Xenbase (<https://www.xenbase.org/xenbase/>). The fixed value peptide spectral match metric was used to estimate the false discovery rate. Identified proteins were semi-quantified based on normalized peptide spectral match counts to approximate relative abundance between samples.

For lipid mediator analysis, the concentration of cysteinyl leukotrienes found in the mast cell degranulates were measured using a cysteinyl leukotriene ELISA kit (Cayman Chemical) following the manufacturer's instructions. Degranulates collected from 2.0×10^6 BMMCs and PMCs pooled from 6 individual animals were combined and assayed in triplicates.

Results

Frog PMCs and BMMCs exhibit canonical mast cell phenotype

To investigate whether hallmark mammalian mast cell subtypes are present in amphibians, we generated *X. laevis* frog

BMMCs and PMCs by culturing source specific precursors with rSCF. BMMCs and PMCs were compared with frog recombinant colony-stimulating factor-3 (rCSF3/recombinant granulocyte colony-stimulating factor)-treated neutrophils, to further confirm mast cell characteristics. Both BMMCs and PMCs possessed large numbers of acidic cytoplasmic granules, whereas neutrophils exhibited canonical multilobed nuclei and no distinct granulation (Fig. 1A). The mast cell-specific carboxypeptidase-3 (CPA3)^{50,51} is a reliable marker of mast cells across vertebrates,^{50,51} and our in situ hybridization analysis confirmed that both frog BMMCs and PMCs, but not neutrophils, expressed CPA3 (Fig. 1B). Notably, CPA3 protein sequences, including those of *X. laevis*, human and mouse, are highly conserved (Fig. S2), indicating that this protein is a reliable marker for mast cells across and outside of mammals.

Mammalian connective tissue mast cell granules contain heparin, while mammalian mucosal mast cell granules are thought to be devoid of heparin.^{52–54} Conversely, both frog BMMCs and PMCs, possessed heparin in their granules (Fig. 1C). Frog neutrophils were entirely absent of heparin-containing granules (Fig. 1C).

Frog PMCs and BMMCs possess canonical mast cell transcriptomic profiles

To date, the functions of nonmammalian mast cells remain largely unexplored. Therefore, to characterize possible frog mast cell subtypes and to delineate functional differences between distinct frog granulocyte subsets, we compared the transcriptional profiles of *X. laevis* BMMCs and PMCs to frog neutrophils and to each other. Our RNA sequencing analysis revealed many differentially expressed (DEGs) immune and nonimmune genes between the 3 granulocyte populations.

Mammalian mast cells comprise distinct tissue-specific subtypes, characterized by the expression of specific proteases.^{52,55,56} The gene expression profiles of PMCs and BMMCs were compared to assess whether they represent transcriptionally distinct mast cell populations. While many of the DEGs did not clearly associate with known functional differences, several of the top DEGs in PMCs were cathepsin genes (Fig. 2A). Cathepsins are lysosomal proteases involved in antigen processing and tissue remodeling, and some are selectively expressed in mast cell subsets, such as cathepsin G in CTMCs but not in MMCs.^{57,58} The PMCs also had high levels of *il7r.s*, which in mammals can form heterodimers with either the common cytokine γ chain, to form the IL-7 receptor, or with the thymic stromal lymphopoietin receptor, to form a high-affinity thymic stromal lymphopoietin receptor.⁵⁹ The BMMCs had high expression of *cd9.L*, which has been implicated in IL-16-mediated mast cell chemotaxis and forming interactions with FcεRI.^{60,61}

In order to further confirm that the PMC and BMMC cultures represented distinct granulocyte populations, their transcriptomes were compared with neutrophils. Many of the genes expressed at greater levels by both frog PMCs and BMMCs when compared with frog neutrophils, encode markers associated with mammalian mast cells. For example, compared with neutrophils, PMCs showed greater expression of *kit. S*, *gata1.S*, *gata2.S*, *ctsh. L*, and *ctsl. L* (Fig. 2B), and BMMCs likewise had greater expression levels of *kit. L/S*, *gata1.L/S*, and *gata2.L/S* (Fig. 2C). Notably, *gzmh. S*, a gene associated with mammalian cytotoxic cells,⁶² was among the

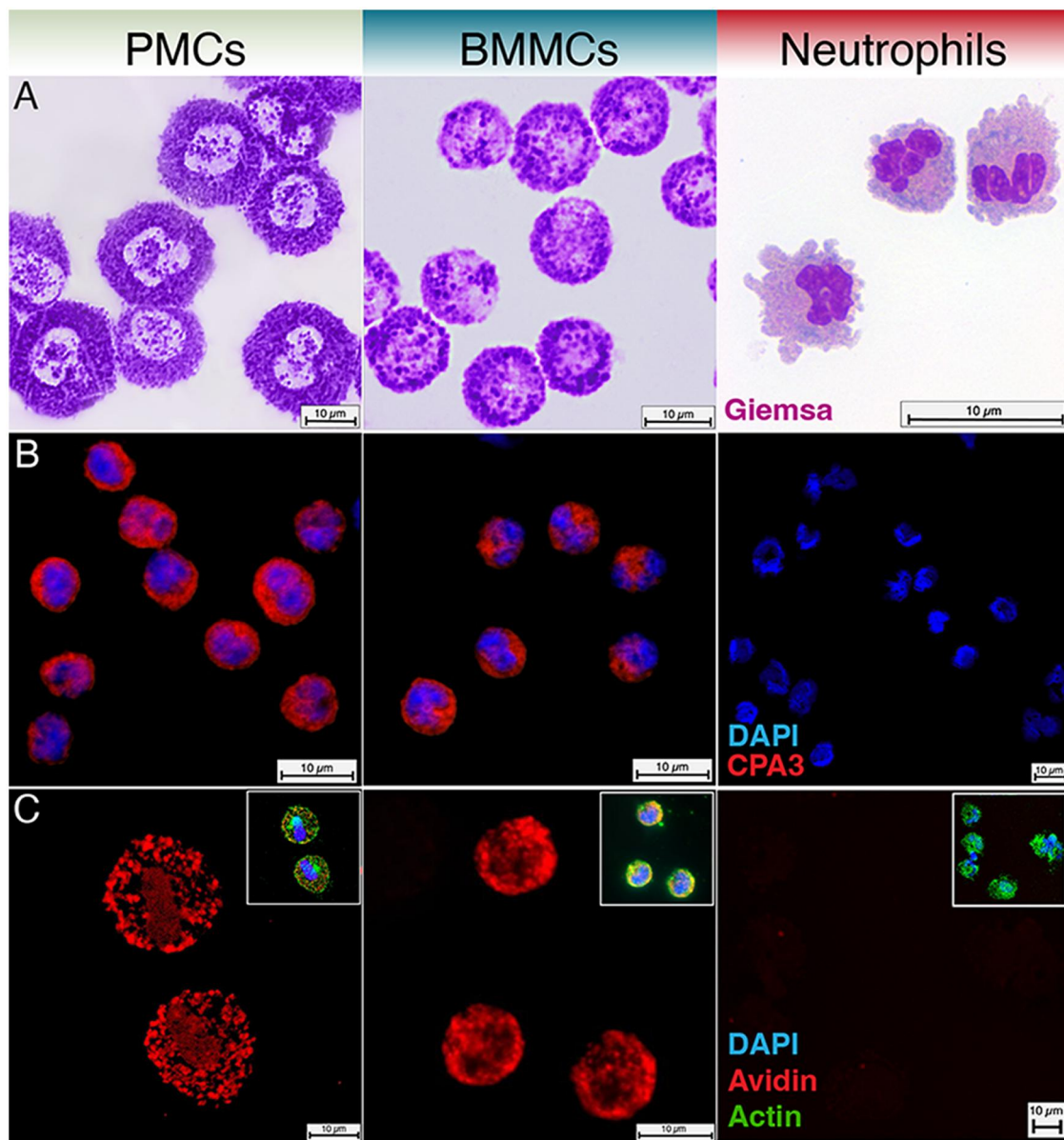


Figure 1. *Xenopus laevis* bone marrow and peritoneal precursor cells cultured with rSCF exhibit characteristic mast cell morphology and express a mast cell-specific protease. Mast cell cultures were generated from bone marrow (BMMCs) or peritoneal (PMCs) precursors and treated with rSCF. Neutrophils were derived from bone marrow precursors and treated with recombinant granulocyte colony-stimulating factor (rGCSF). **(A)** Giemsa staining of PMCs and BMMCs shows canonical mast cell morphology, distinct from cultured neutrophils, which have multilobed nuclei. **(B)** PMCs and BMMCs express the mast cell-specific CPA3, which was not detected in neutrophils. CPA3 messenger RNA was visualized by in situ hybridization using RNAscope. **(C)** PMCs and BMMCs contain intracellular heparin-positive granules, detected using Avidin-TxRed. Neutrophils lacked heparin staining. False coloring was applied to grayscale Giemsa cytology images (Fig. S1).

most highly expressed genes in both PMC and BMMC populations relative to neutrophils (Fig 2B, C). Although not yet functionally characterized, the *X. laevis* KIT-SCF receptor possesses hallmark features of other vertebrate KIT proteins, including 5 N-terminal Ig domains and C-terminal discontinuous kinase domains (Fig. S3),⁶³ The high expression of this gene in frog mast cells confirms that akin to mammals, KIT is likewise crucial to frog mast cell biology. This is intuitive, considering that rSCF/KIT-ligand was used to generate these cultures.

When compared with PMCs, neutrophils exhibited significantly higher expression of canonical neutrophil associated genes such as *cxc4.L*, *mmp9.1.S*, *mpo.L*, and *lta4b.L* (Fig. 2B). Neutrophils also possessed a greater level of

transcripts for *mmp9.1.S*, *lta4b.L*, *ncf1.L*, and *ncf2.L* than BMMCs (Fig. 2C).

Principal component (PC) analysis of the normalized messenger RNA expression data indicated that PMCs and BMMCs clustered closely together, while neutrophils formed a distinct group. PC1 explained 41% of the variability in gene expression across the 3 granulocyte populations, making it the most informative factor for distinguishing among the cell types. PC2 accounted for an additional 15% of the variability. Together, PC1 and PC2 captured the majority of the variance in the dataset, enabling visualization of the transcriptional similarities and differences between the granulocyte populations (Fig. 2D). Additionally, there was substantial overlap between the DEGs identified in PMCs

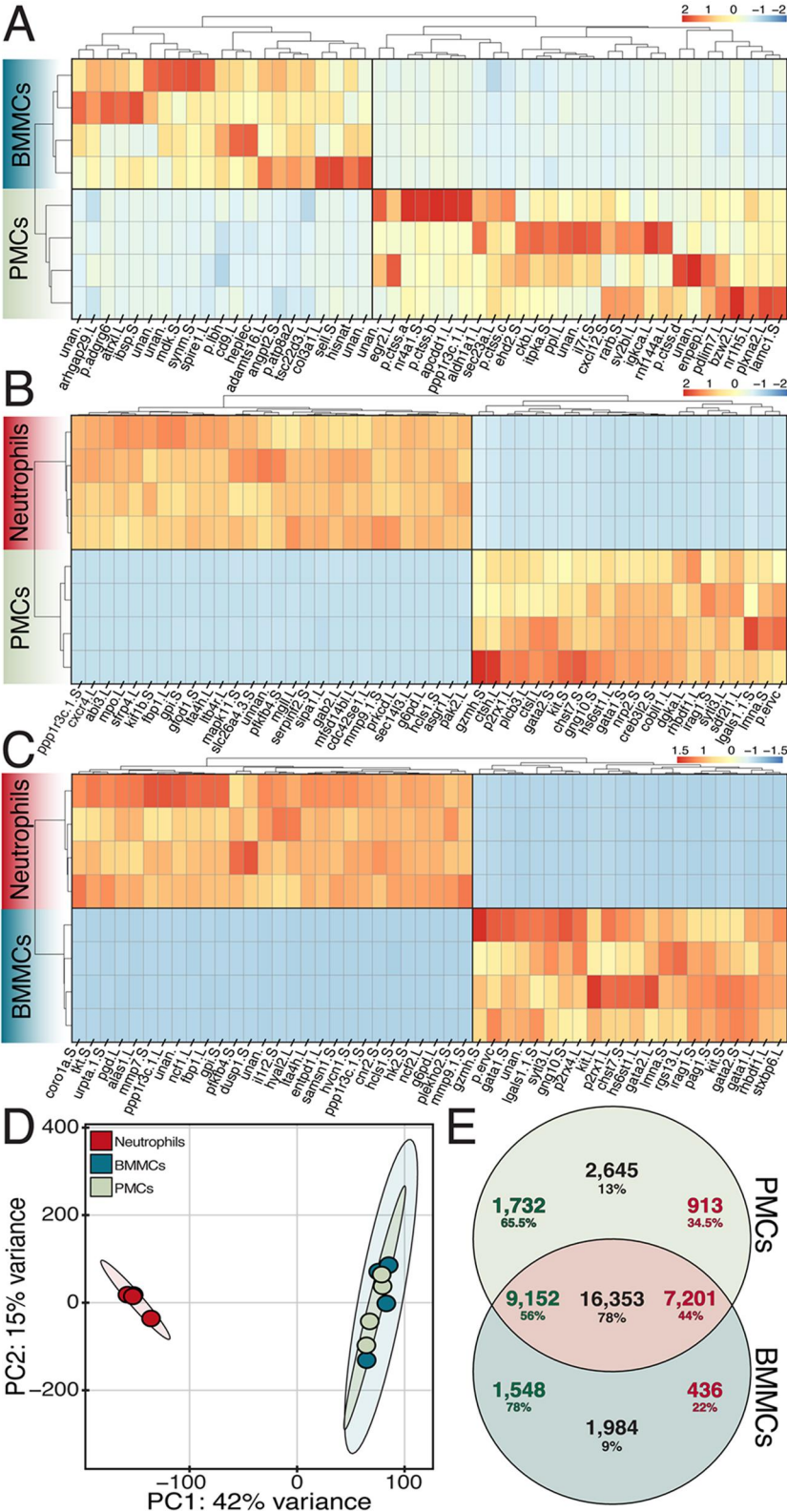


Figure 2. The transcriptomic profiles of frog PMCs and BMMCs reveal canonical mast cell gene expression and are distinct from those of neutrophils. Frog BMMC-, PMC-, and neutrophil-derived RNA was analyzed by bulk RNA sequencing. DEGs were identified using the DESeq2 pipeline in R. **(A)** The top 50 DEGs between PMCs and BMMCs. **(B)** Top 50 DEGs between PMCs and neutrophils. **(C)** The top 50 DEGs between BMMCs and neutrophils. Genes that are unannotated or lack strong homology to known genes are labeled as “unan.” and the corresponding locus identifiers are listed in the supplemental materials. **(D)** PC analysis of normalized gene expression across PMCs, BMMCs, and neutrophils. Ellipses represent 95% confidence intervals for each cell type. **(E)** Venn diagram showing overlap of DEGs in PMCs and BMMCs relative to neutrophils. Total DEGs are shown in black, with upregulated genes in green and downregulated genes in red.

versus neutrophils and those found in BMMCs versus neutrophils. Of the total DEGs, 78% were shared between PMCs and BMMCs, with 56% upregulated and 44% downregulated in both cell types. Each cell type also exhibited a comparable number of unique DEGs, with PMCs and BMMCs having 13% and 9% unique genes, respectively (Fig. 2E).

Frog PMCs, BMMCs, and neutrophils differ in their expression of immune genes

The previous transcriptomic analysis provided an overview of global gene expression differences between frog granulocyte types. However, many immune effector genes were not captured, limiting inferences about potential functional differences. To further explore potential differences between frog BMMCs and PMCs, as well as functional distinctions from neutrophils, we analyzed subsets of hallmark immune-related genes from our transcriptomic datasets.

Mammalian mast cell subtypes differ in their repertoires of proteases.⁶⁴ To evaluate potential differences between PMCs and BMMCs, and to distinguish them from neutrophils, we assessed possible differences in protease gene expression. PMCs and BMMCs showed relatively high expression of *cpa3.L*, putative (p) *cpa3* (*p.cpa3*), and *gzmb.L/S* (Fig. 3A), consistent with mast cell identity and supporting earlier cytological analysis of *cpa3* expression (Fig. 1B). However, the PMCs had relatively higher expression of *gzma.L/S* and *gzmb.L* compared with BMMCs, highlighting some differences in protease repertoire between these mast cell types. Chymase, a protease associated with connective tissue mast cells, has not been identified in *X. laevis*.⁶⁴ However, the putative (p.) chymase gene (*p.chymase*) did not show a clear expression difference between PMCs and BMMCs, and was instead most highly expressed by neutrophils (Fig. 3A). Of the proteases examined, *cpa3.L*, *p.cpa3*, and *gzmb.S* were the most highly expressed in both PMCs and BMMCs. In contrast, neutrophils exhibited low or no expression of these proteases, with the exception of *prss27* (serine protease 27) (Fig. 3A).

When activated, mammalian mast cells release a variety of preformed molecules, including inflammatory proteins and lipid mediators such as leukotrienes.⁶⁵ The frog PMCs and BMMCs exhibited high expression of *lta4s1.L*, while neutrophils showed high expression of *lta4b.L*, suggesting that frog granulocytes are likewise capable of lipid mediator production. Of note, neutrophils expressed significantly higher levels of all leukotriene receptors examined (Fig. 3B), consistent with mammalian systems in which neutrophils are chemoattracted to sites of infection by leukotrienes.⁶⁶

Mast cells are tissue resident immune cells and function as the first responders to infection, typically mediated through pattern recognition receptors such as TLRs.⁶⁷ To evaluate the range of pathogen recognition capabilities, we compared the expression of TLRs among the frog granulocytes. PMCs and BMMCs had greater expression of *tlr2.L/S*, *tlr3.L*, *tlr4.S*, and *tlr6.L/S*, with the PMCs having greater expression of *tlr9.L*. BMMCs possessed greater *tlr2.S* transcript levels and neutrophils had greater expression of *tlr5.L* (cell surface TLR5) and *tlr22.S*. All 3 granulocyte types possessed comparable expression of *s.tlr5.L* (soluble TLR5) (Fig. 3C).

With respect to chemokine ligand and receptor expression, BMMCs and PMCs possessed significantly higher expression of *ccl42d.L*, *ccl42a.S*, *cxcl2.L*, *p.cxcl8.b*, *cxcl16.L/S*, and *cxcl1.S* compared with neutrophils (Fig. 4A). PMCs had

greater expression of *ccl5.L*, *ccl19.L*, and *cxcl12.S*, while BMMCs showed higher transcript levels of *cxcl14.L* (Fig. 4A). Neutrophils possessed lower expression of all chemokine ligand genes but expressed greater levels of the chemokine receptor genes, *cxcr1.L/S* and *cxcr4.L* (Fig. 4B), both genes being hallmarks of mammalian neutrophils, with the former also being key to frog neutrophil biology.^{68–70} Both BMMCs and PMCs had greater expression of *ccr2.L*, *ccr7.L*, and *ackr4.L* (Fig. 4B).

Mast cells play a key role in type 2 immune responses, particularly in defense against parasites.⁷¹ Upon activation, they release both preformed and newly synthesized cytokines that help alert the immune system and elicit the appropriate cellular response.⁷² While the cytokine ligands and receptors involved in type 2 immunity are well characterized in mammals, much less is known about the corresponding effector molecules and cell types involved in amphibian type 2 immune responses. Therefore, we next examined the expression of genes encoding cytokines and receptors. Compared with neutrophils, both mast cell populations showed elevated expression of *p.il12 β* , *il12 β .S*, *il21.L*, *il15.S*, and *p.il26* relative to neutrophils. Notably, BMMCs expressed significantly greater levels of *il21.L* than PMCs or neutrophils (Fig. 5A). Our past work indicated that frog mast cells are major sources of IL-4.³⁹ Therefore, it was surprising that all 3 granulocyte populations shared low *il4* expression (Fig. 5A). Compared with the MC subsets, neutrophils showed higher expression of *il1 β .S*, *il16.L*, *p.il16*, *p.il18*, and *tgfb1.L/S* (Fig. 5A). Together, both mast cell types predominantly expressed type 1 and 2 cytokines, neutrophils primarily expressed proinflammatory cytokines.

The 3 granulocyte types exhibited greater diversity in cytokine receptor expression than cytokine ligand expression. PMCs possessed greatest expression of *il7R.L/S*, BMMCs showed highest *p.il17Re* expression while both PMCs and BMMCs expressed greater levels of *il17Re* compared with neutrophils (Fig. 5B). Both mast cell types had high expression of *p.il12R β .c*, *p.il2R β .b*, *il2R γ .L*, *il15Ra.L/S*, *ifn γ R1.S*, *ifn γ R2.2.S*, *il4R.L/S*, *p.il13Ra1.a*, *il13Ra2.L*, *il21R.S*, *p.il17Ra.a*, *il17Rc.L*, *il1R1.S*, *ifnaR1.S*, *ifnaR2.L/S*, and *tgfb2.L* (Fig. 5B). Notably, both mast cell types expressed greater levels of *il2R γ .L*, which in mammals encodes the common cytokine receptor γ -chain that forms heterodimers with IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21 receptors.⁷³ Thus, the elevated expression of this gene in mast cells is consistent with these cells also expressing greater levels of *p.il2R β .b*, *il15Ra.L/S*, and *il4R.L/S*. Conversely, frog neutrophils had greater expression of *p.ifn γ R2*, *p.il5Ra.a*, *p.il5Ra.b*, *p.il5Ra.c*, *p.il13Ra1.b*, *il17Ra.S*, *p.il1R1.b*, *il1R2.L*, *il18R1.L*, *p.il18R1*, *il10R β .S*, and *tgfb2.S* (Fig. 5B).

Frog PMCs and BMMCs possess robust degranulation capacities

Our cytological and transcriptomic comparisons clearly distinguished BMMCs and PMCs from neutrophils but less so from each other. Many mast cell mediators are preformed and stored in granules rather than actively transcribed, while others are synthesized as peptides and may not be detectable at transcript levels.¹⁹ To evaluate both preformed and de novo synthesized mediators, we conducted degranulation assays, mass spectrometry, and a leukotriene ELISA, to discern BMMC and PMC degranulation capacities and contents. Using canonical degranulation assays, measuring

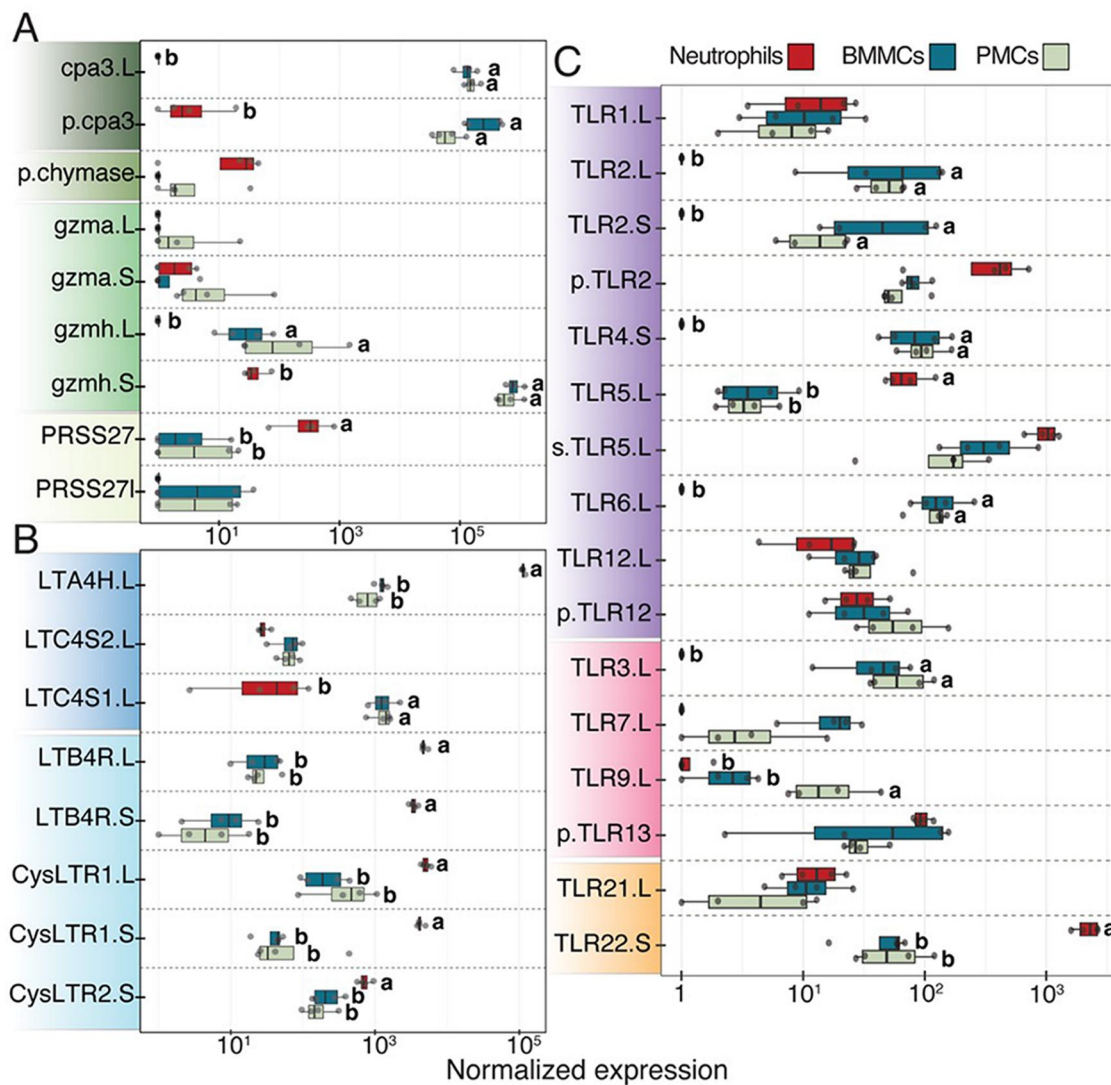


Figure 3. PMCs, BMMCs, and neutrophils display distinct expression of proteases, leukotriene synthesis enzymes, leukotriene receptors, and TLRs. Normalized read counts for selected protease, leukotriene, and TLR genes were compared among PMCs, BMMCs, and neutrophils. **(A)** Expression of mast cell-associated proteases. Green shading distinguishes protease families (carboxypeptidases, chymase, granzymes, and serine proteases). **(B)** Expression of leukotriene-synthesizing enzymes (dark blue) and leukotriene receptors (light blue). **(C)** Expression of cell surface TLRs (purple), endosomal TLRs (pink), and nonmammalian TLRs (orange). Each cell type is represented by 4 biological replicates (gray points), and the average expression is shown as a black bar within colored boxplots. Letters indicate genes with statistically significant differences, with different letters representing significant difference in gene expression. Statistical significance was determined using pairwise Wilcoxon rank-sum tests with P values adjusted for multiple comparisons using the false discovery rate. Adjusted $P < 0.05$.

β -hexosaminidase and tryptase activity, we showed that PMCs and BMMCs possessed comparable capacities to degranulate in response to calcium ionophore (A23187) stimulation (Fig. 6A), confirming for the first time that frog MCs mast cells are capable of degranulation and that both enzymes are conserved in amphibian MCs. Additionally, PMCs treated with A23187 exhibited a reduction in heparin-positive granules as well as detection of granules budding from the cell surface, further confirming through cytology the ability of these cells to degranulate (Fig. S6).

Proteomic analysis confirmed that many of the more abundant degranulate contents were shared between BMMCs and PMCs, including cathepsin K, ervatamin C-like (cysteinyl protease found in *Ervatamia coronaria*), carboxypeptidase B-like (CPB-like), and granzyme H (Fig. 6B).⁷⁴ Notably, some effectors were uniquely detected in one MC type and not the another. PMC degranulates contained HSP70, calmodulin,

and galectin, whereas BMMC degranulates contained carboxypeptidase B, DNase II, and dipeptidyl peptidase 1 (Fig. 6B).

Proteases, glycosidases, and receptors were the most abundant functional categories identified in both cell types. However, the distribution of these molecules varied, with BMMCs showing a higher number of glycosidases and PMCs having a greater number of immune-related proteins such as galectin, granulin, tachylectin-like proteins, and lysozyme (Fig. 6C).

To discern potential differences between lipid mediator content in BMMC and PMC degranulates, we performed a leukotriene ELISA. We detected cysteinyl leukotrienes in the degranulate from both mast cell cultures, PMCs exhibiting higher concentrations of cysteinyl leukotrienes in their degranulates (14.2 ng/mL) compared with BMMCs (6.4 ng/mL) (Fig. 6D).

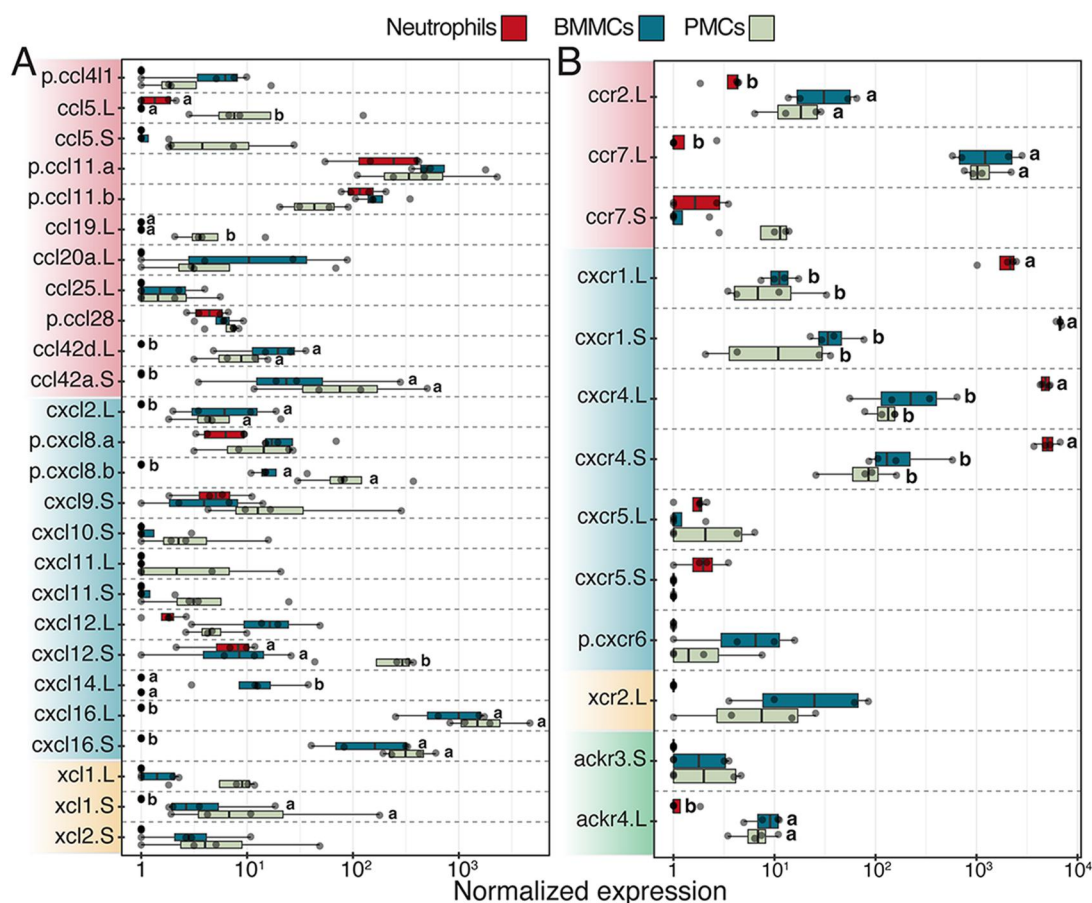


Figure 4. PMCs, BMMCs, and neutrophils exhibit distinct expression profiles of chemokine ligands and chemokine receptors. Normalized read counts for selected chemokine ligands and chemokine receptors. **(A)** Expression of chemokine ligands. **(B)** Expression of chemokine receptors. Colored gradients organize genes by chemokine family: red for CC, blue for CXC, yellow for C, and green for ACKR. Each cell type is represented by 4 biological replicates (gray points), and the average expression is shown as a black bar within colored boxplots. Letters indicate genes with statistically significant differences, with different letters representing significant difference in gene expression. Statistical significance was determined using pairwise Wilcoxon rank sum tests with *P* values adjusted for multiple comparisons using the false discovery rate. Adjusted *P* < 0.05.

Discussion

Our findings indicate that frog PMCs and BMMCs share many features associated with mammalian mast cells. Frog PMCs and BMMCs possess acidic granules containing heparin and express mast cell CPA3, β -hexosaminidase, tryptase, and cysteinyl leukotrienes. Our work also indicates that akin to mammals, amphibians possess mast cell precursors in their peritoneal cavities and bone marrow. Thus, functional mast cell equivalents resembling those found in mammals were present at least 200 million years ago, when anuran amphibians first emerged.⁷⁵ Nearly all of what is known about mast cell biology has been established using mammalian models. However, mast-like cells are found across all vertebrate lineages, and cells with properties of mast cells have even been identified in invertebrate deuterostomes such as ascidians,^{31,76,77} suggesting that cells bearing at least some mast cell properties could be traced to animals that emerged 500 million years ago. To our knowledge, mast-like cells have not been described in jawless and cartilaginous fishes. In bony fish, mast-like cell effector responses are influenced by helminth infections,⁷⁸ alongside our recent frog work,³⁹ highlighting that mast cells and type 2 immune responses evolved prior to mammals. Amphibian mast (like) cells have been described prior to our studies, and in fact the first mast-like cells were described in 1863 in frogs.^{1,39,79–82} As

amphibians occupy a pivotal position in vertebrate evolution, being the first vertebrates to transition to terrestrial life, they represent an informative model for studying the evolution of the vertebrate immune systems, including mast cell biology.^{83,84} Here we successfully culture mast cells from peritoneal-derived and bone marrow-derived precursors, examine their degranulation capacity, and characterize the resulting degranulate through proteomic and leukotriene analyses in a nonmammalian system. Collectively, these findings detail and compare the immune effector profiles of 2 frog mast cell cultures and offer new insights into the evolution of mast cell-mediated immune responses across vertebrates.

Mammalian mast cells reside in tissues that interface with external environments, including mucosal surfaces such as the respiratory and gastrointestinal tracts, as well as connective tissues such as the skin and smooth muscle.^{3,14} Due to their strategic tissue localization, mast cells serve as integral sentinels of the immune system. They are often described as the “loaded gun” of the immune system due to their abundance of preformed bioactive mediators within cytoplasmic granules, which are rapidly released upon detection of perceived immune threats.^{85,86} Our findings confirm that many of these attributes are conserved in frog mast cells, which express and synthesize chemokines, cytokines, proteases, and leukotrienes. This is exemplified by the analysis of

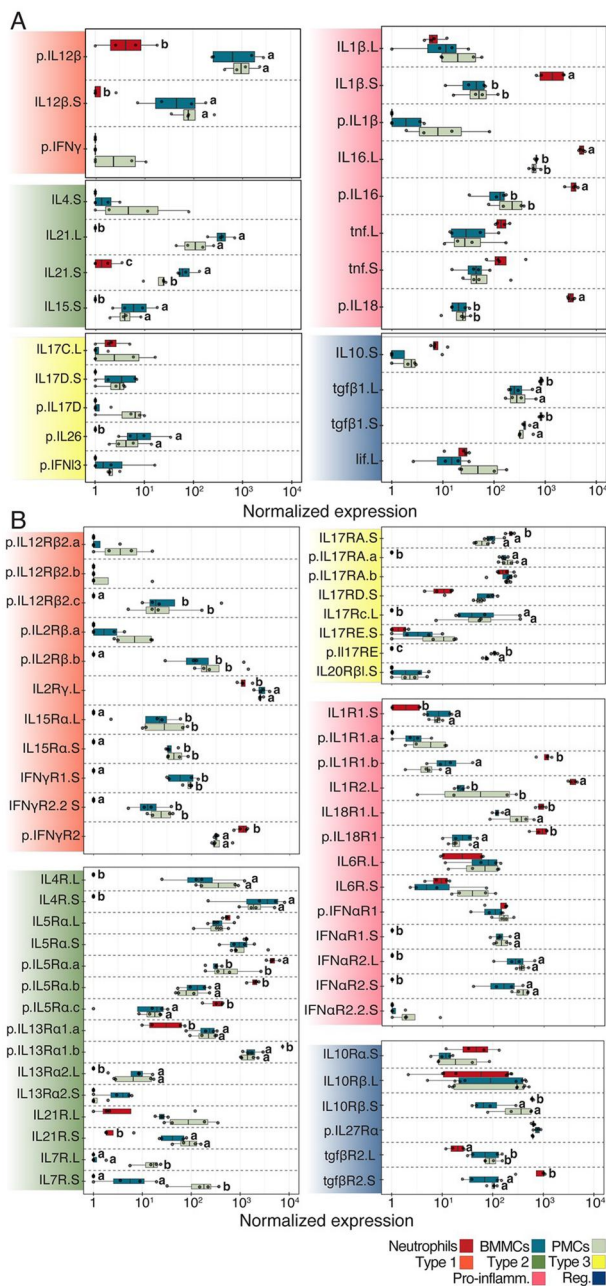


Figure 5. PMCs, BMMCs, and neutrophils exhibit distinct expression profiles of cytokines and cytokine receptors. Normalized read counts for selected cytokine ligands and cytokine receptors. **(A)** Expression of cytokine ligands. **(B)** Expression of cytokine receptors. Colored gradients organize genes by general cytokine function: orange for type 1, green for type 2, yellow for type 3, pink for proinflammatory, and blue for regulatory. Each cell type is represented by 4 biological replicates (gray points), and the average expression is shown as a black bar within colored boxplots. Letters indicate genes with statistically significant differences, with different letters representing significant difference in gene expression. Statistical significance was determined using pairwise Wilcoxon rank sum tests with *P* values adjusted for multiple comparisons using the false discovery rate. Adjusted *P* < 0.05.

degranulates from both PMCs and BMMCs, which contained potent proinflammatory mediators, proteases, and immune proteins. However, other bioactive molecules such as AMPs, which we anticipated to find in frog mast cell degranulates, were not identified. Previous studies in teleost fish demonstrated that mast-like cells contain AMPs, which influence bacterial infections.⁷⁸ Because amphibians rely heavily on

AMPs to combat infections, particularly in the skin,⁸⁷ further analyses should assess whether amphibian mast cells are capable of releasing AMPs, perhaps under distinct conditions.

A hallmark feature associated with mammalian mast cells is their rapid release of histamine, although mast cell-produced lipid mediators are far more biologically active than histamine and confer numerous immune roles associated with mast cells.^{88,89} The mast cells of most fish and amphibians are presently not thought to produce histamine while our results clearly indicate that frog mast cells are capable of lipid mediator responses, including expressing enzymes associated with leukotriene biosynthesis and release of cysteinyl leukotrienes.⁸¹ It will be interesting to learn with future studies how nonmammalian mast cells utilize histamine (if at all) and respective lipid mediator responses to confer distinct outcomes, compared with mammalian successful and aberrant mast cell responses.

Mammalian mast cells differ depending on their tissue residence, with 2 major subtypes recognized as either CTMCs or MMCs.⁹⁰ Although these cells are functionally similar, their tissue-specific environments also dictate some unique characteristics. For example, CTMCs are associated with a diverse repertoire of proteases including chymase, tryptase, and cathepsin G, while the major protease in MMCs granules is tryptase. Additionally, the granules of CTMCs contain the proteoglycan heparin, which can be detected using avidin (as employed here) and allows for distinguishing CTMCs and MMCs.^{26,54,91,92} These distinct mast cell subtypes are also derived from distinct precursors. The relatively long-lived CTMCs are replenished by tissue-specific myeloid progenitors, seeded into respective tissues early on in development. Conversely, MMCs are replaced by circulating hematopoietic stem cells, have a shorter lifespan and thus require steady state repopulation.⁹³ These mammalian mast cell subtypes are thought to require distinct, tissue microenvironment-specific cytokines for differentiation.²⁶ CTMCs development is associated with fibroblast derived cytokines such as SCF and IL-4.²⁹ While MMCs development requires T cell derived cytokines such as TGF- β , IL-3, and IL-9.⁹⁴ Various protocols have been used to establish the different mast cell subtypes in vitro, but they typically involve specific combinations of cytokines and precursor sources. CTMCs are most often generated from peritoneal-derived precursors, treated with SCF, IL-3, and IL-4, while MMCs are commonly derived from bone marrow precursors cultured with SCF, IL-3, IL-9, and TGF- β .^{29,95–97} Notably, adoptive transfer experiments indicate that these mammalian mast cell subtype phenotypes are plastic, as CTMCs transplanted into a mucosal environment can transition into MMC-like, and vice versa.⁹⁸

Extrapolating mammalian mast cell subtype characteristics to amphibians is complicated by fundamental differences in life history, hematopoiesis, and tissue organization. For example, while mammalian CTMCs are typically identified by heparin staining, we previously observed that mast cells in amphibian skin can also be visualized with this marker.³⁹ This is particularly interesting because amphibian skin serves as both a mucosal and connective tissue barrier, containing secretory and ciliated epithelial cells like mammalian mucosal epithelia.⁹⁹ Thus, the presence of heparin-positive mast cells in amphibian skin reveals unique aspects of mast cell biology across tissues that do not correspond to mammalian categories. Additionally, our results show that both PMCs and BMMCs are heparin positive, and therefore heparin may not

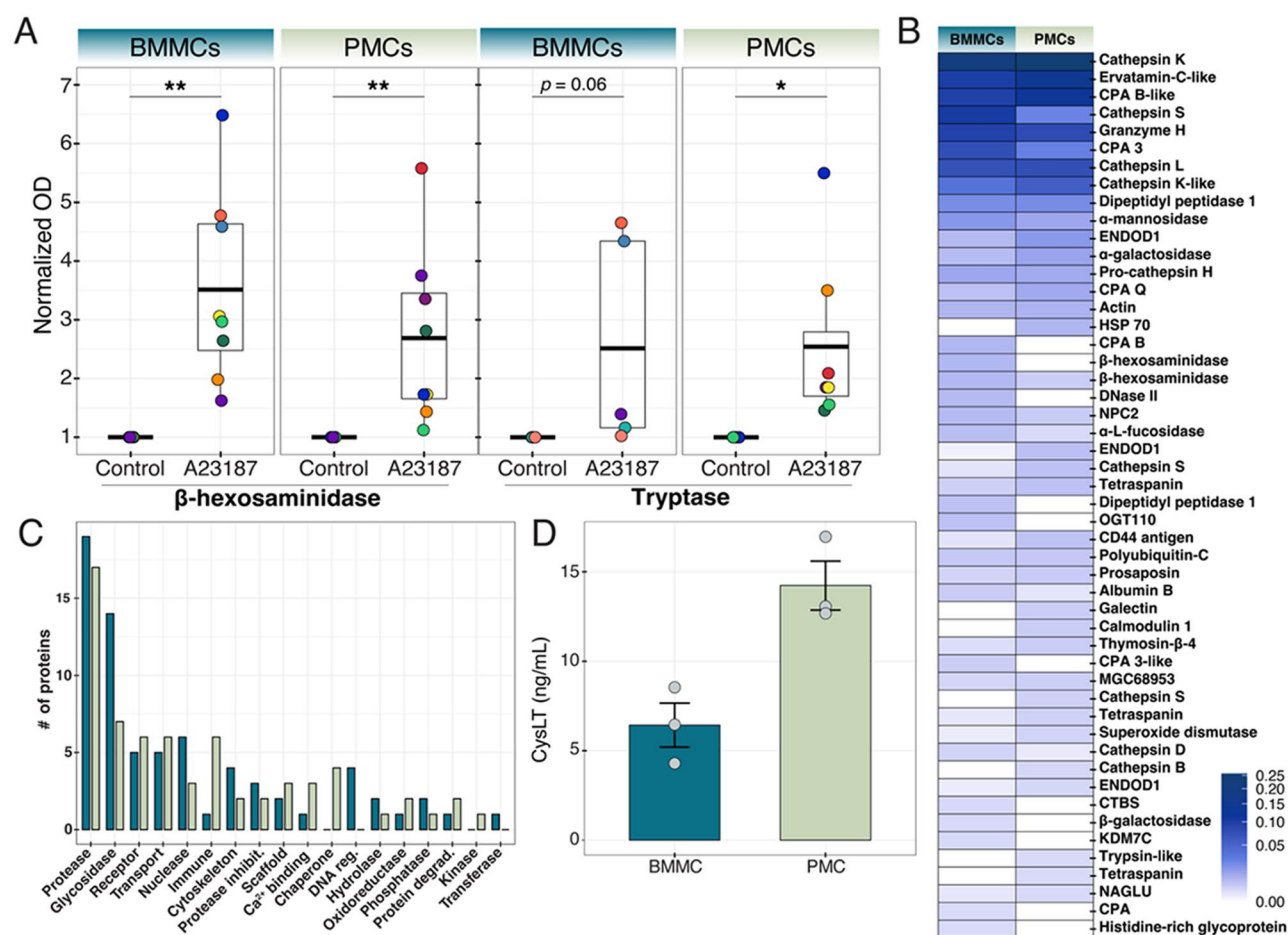


Figure 6. PMCs and BMMCs undergo degranulation and release preformed mediators including a wide array of proteases, glycosidases, and at least 1 type of leukotriene. PMCs and BMMCs were stimulated with calcium ionophore (A23187), and degranulation was confirmed through enzymatic assays and cytological analysis. Degranulate from both mast cell cultures were analyzed by proteomics and a leukotriene ELISA. **(A)** Enzymatic activity of β -hexosaminidase and tryptase from A23187 stimulated PMCs and BMMCs, measured at an optical density (OD) of 405 nm and normalized to control values. Colored dots represent individual replicates of mast cells cultured from different animals, and the black bar within each boxplot indicates the average normalized OD. Statistical significance was determined using a paired Wilcoxon signed rank test. $*P < 0.05$; $**P < 0.01$. **(B)** Heatmap of proteins identified in PMC and BMMC degranulate. Color intensity reflects relative abundance, estimated using normalized peptide spectral match (PSM) counts from mass spectrometry. **(C)** Number of proteins grouped by functional category identified in PMC and BMMC degranulates. Functional groups were manually curated. Any keratins were excluded from analysis. **(D)** Concentration of cysteinyl leukotrienes (CysLTs) detected in PMC and BMMC degranulate. Each grey dot represents a technical replicate from pooled degranulate derived from 2 million cells.

the best marker for distinguishing between mast cell subtypes outside of mammals. In mammals, cytokines such as IL-3 are key drivers of mast cell subtype differentiation, particularly for mucosal mast cells.⁹⁴ Although amphibian IL-3 has not been identified, the presence of an IL-3 receptor gene suggests that this pathway may be conserved in frogs. Indeed, both PMCs and BMMCs exhibit relatively high expression of the common cytokine γ -chain, which in mammals forms a heterodimer with the IL-3 receptor, raising the possibility that a related signaling mechanisms influence mast cell differentiation in amphibians.⁷³ Different amphibian species also markedly differ in where their hematopoiesis takes place.¹⁰⁰ In *X. laevis*, megakaryocyte/erythroid precursors mature in the liver periphery, whereas granulocyte/macrophage precursors are localized within the bone marrow.^{101,102} Amphibians also undergo metamorphosis, which profoundly reshapes their immune systems.¹⁰³ As such, the development and maturation of distinct amphibian mast cell subsets is likely at least in part different from mammals and may be further dictated by amphibian orders, genera, and species.

While PMCs and BMMCs shared many core features, differences in their transcriptomic profiles and degranulate

proteome composition, particularly the presence of unique proteases expressed in PMCs, suggest the existence of functionally distinct mast cell subsets in amphibians. For example, when comparing the transcriptomic profiles between PMCs and BMMCs, the PMCs had significantly higher expression of cathepsin-S related genes. Albeit not functionally characterized, these proteases may represent distinguishing features of frog mast cell subsets, akin to CTMC-specific chymase production.¹⁰⁴ Moreover, our findings indicate that frog PMCs and BMMCs express disparate chemokine ligands, suggesting targeted recruitment of distinct leukocyte populations by these frog subsets. These differences suggest the presence of distinct mast cell subsets in amphibians, although these subsets may differ from the mammalian connective tissue and mucosal mast cell subsets.

We previously showed that *X. laevis* mast-like cells could be generated as short-term cultures from bone marrow precursors. While these cells expressed several mast cell markers, including transcription factors (*gata1*, *gata2*) and the receptor tyrosine kinase *kit*, cytological and transcriptomic analyses suggested they were not fully mature,³⁹ an observation that has been confirmed by the present studies. Interestingly,

enriching frog skin mast cells conferred greater anti-fungal protection against the *Batrachochytrium dendrobatidis* chytrid fungus, underscoring the importance of these cells to amphibian skin immune defenses.³⁹ Our past studies suggest that antifungal protection conferred by frog skin mast cells is at least in part mediated by the type 2 cytokine, IL-4. Interestingly, PMCs and BMMCs exhibited relatively low *il4*. *L/S* expression, suggesting that at least in vitro, amphibian mast cells do not produce copious amounts of this cytokine. Instead, frog mast cell *il4* expression is presumably upregulated in response to tissue microenvironments and immune activation. The BMMC and PMC cultures described here represent pertinent platforms upon which to further discern the precise relationships between frog mast cells and IL-4.

Our findings offer new insights into the biology of nonmammalian mast cells and how these features relate to canonical (mammalian) mast characteristics. Mammalian and chiefly human mast cells are undervalued for their immunological roles and villainized for their key involvement in diseases such as allergies, which are characterized by mast cell mediators like histamine and degranulation in response to IgE.¹⁰⁵ Of note, histamine has yet to be identified in frogs,⁸¹ and amphibians lack IgE, instead encoding the IgY predecessor to both IgG and IgE.^{81,106} While our findings confirm that many features of mammalian mast cells are conserved in amphibians, we also identified instances of divergence, with additional differences from mammalian mast cells likely remaining to be discovered. Through alternative animal models such as amphibians, and in vitro systems such as those described here, we can achieve unique perspectives on the evolution of mast cell biology in context of vertebrate health and disease.

Author contributions

R.C.: Conceptualization, data curation, formal analysis, investigation, methodology, validation, visualization, writing—original draft, writing—review & editing. H.T.: conceptualization, data curation, formal analysis, investigation, methodology, validation, visualization, writing—original draft, writing—review & editing. E.R.C.: data curation, formal analysis, investigation, methodology, validation, visualization, writing—original draft. C.N.G.G.: data curation, formal analysis, investigation, methodology, validation, visualization, writing—original draft. M.R.H.H.: data curation, formal analysis, investigation, methodology, validation, visualization, writing—original draft. J.M.: data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing—original draft. L.G.: Conceptualization, data curation, formal analysis, funding acquisition, investigation, project administration, resources, methodology, validation, visualization, writing—original draft, writing—review & editing.

R.C. (Conceptualization [Lead], Data curation [Lead], Formal analysis [Lead], Investigation [Lead], Methodology [Lead], Validation [Lead], Visualization [Lead], Writing—original draft [Lead], Writing—review & editing [Lead]), H. T. (Conceptualization [Equal], Data curation [Equal], Formal analysis [Equal], Investigation [Equal], Methodology [Equal], Validation [Equal], Visualization [Equal], Writing—original draft [Equal]), E.R.C. (Data curation [Equal], Formal analysis [Equal], Investigation [Equal], Methodology [Equal], Validation [Equal], Visualization [Equal], Writing—original

draft [Equal]), C.N.G.G. (Data curation [Equal], Formal analysis [Equal], Investigation [Equal], Methodology [Equal], Validation [Equal], Visualization [Equal], Writing—original draft [Equal]), M.R.H.H. (Data curation [Equal], Formal analysis [Equal], Investigation [Equal], Methodology [Equal], Validation [Equal], Visualization [Equal], Writing—original draft [Equal]), J.M. (Data curation [Equal], Formal analysis [Equal], Investigation [Equal], Methodology [Equal], Software [Equal], Validation [Equal], Visualization [Equal], Writing—original draft [Equal]), and L.G. (Conceptualization [Equal], Data curation [Equal], Formal analysis [Equal], Funding acquisition [Lead], Investigation [Equal], Methodology [Equal], Project administration [Lead], Resources [Equal], Supervision [Lead], Validation [Equal], Visualization [Equal], Writing—original draft [Lead], Writing—review & editing [Lead])

Supplementary material

Supplementary material is available at *The Journal of Immunology* online.

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Conflicts of interest

The authors declare no conflicts of interest.

Data availability

Data are available upon request.

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